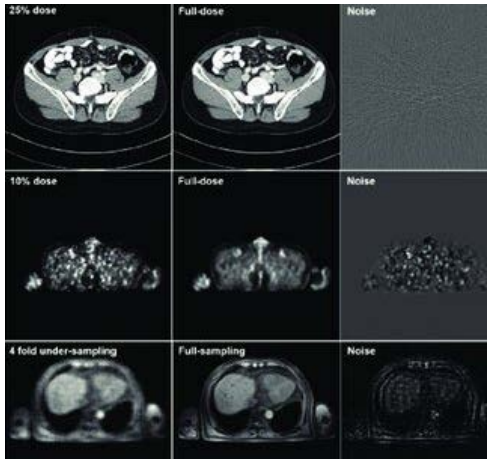
EUCAIM
CANCER IMAGE EUROPE



Dimitris Filos, phd

Imaging data - Presentation of imaging repositories - clinical challenges



dimfilos@auth.gr

Imaging standards/file formats

- DICOM
- NIfTI image standard
- **MetaImage** standard (.mha)
- nrrd
- Open Microscopy Environment TIFF
  - The most important **open standard** for WSI
  - .SVS file format

Medical Imaging File Formats

- Medical imaging uses several file formats, each designed for different needs such as clinical use, research, multidimensional data, or large image archives.
- Widely used formats:
  - **DICOM** – the clinical standard
  - **NIfTI** – neuroimaging standard
  - **NRRD** – flexible research format
  - **MHA (MetaImage)** – ITK/SimpleITK-oriented format
- **Whole Slide Imaging (WSI)**
  - SVS (Leica/Aperio)
  - DICOM WSI (emerging clinical standard)
  - OME-TIFF (open research standard)

DICOM (Digital Imaging and Communications in Medicine)

- **Purpose:** Clinical imaging standard (PACS, hospitals)
  Key Features:
  - Rich metadata (patient, scanner, acquisition, modality)
  - Multi-file structured format (one file per slice commonly)
  - Supports all clinical modalities (CT, MRI, US, PET, WSI pathology)
  - Standard for **whole slide imaging (WSI)**
  - Fully supported by clinical systems and regulatory frameworks
- **Strengths:** Universal, heavily standardized, metadata-rich
- **Weaknesses:** Complex, heavy, not ideal for research pipelines

NIfTI (.nii / .nii.gz)

- **Purpose:** Neuroimaging research (MRI, fMRI, DTI)
- **Key Features:**
  - Compact single-file format
  - Coordinates and orientation encoded in header
  - Supports 3D/4D data (time series)
  - Limited metadata structure
- **Strengths:** Simple, standardized in neuroimaging, widely supported
- **Weaknesses:** Not flexible for non-brain imaging or >4D data

NRRD (Nearly Raw Raster Data)

- **Purpose:** General scientific and medical imaging (flexible N-dimensional data)
- **Key Features:**
  - Text-based header (easy to read)
  - Supports arbitrary dimensions (2D, 3D, 4D, ND)
  - Very flexible metadata
  - Supports compressed encodings (gzip, bzip2)
  - Works well with 3D Slicer, VTK, MITK, ITK
- **Strengths:** Extremely flexible; ideal for research, segmentation, diffusion MRI
- **Weaknesses:** Not standardized clinically; less widespread than DICOM/NIfTI

MHA / MetaImage (.mha / .mhd)

- **Purpose:** Research imaging, especially with ITK/SimpleITK
- **Key Features:**
  - Simple, human-readable header
  - Supports embedded data (MHA) or external raw data (MHD + RAW)
  - Designed for 3D/4D images and labeling masks
  - Very common in algorithm development and scientific workflows
- **Strengths:** Very easy to parse; ideal for prototyping
- **Weaknesses:** Limited metadata; not a clinical standard

- https://commission.europa.eu/topics/public-health/european-health-union/cancer-plan-europe_en
- https://www.youtube.com/watch?v=xeFj-8inRAw
- https://digital-strategy.ec.europa.eu/en/policies/cancer-imaging

European Health Data Space (EHDS): Principles and Vision

EHDS is a European initiative to create a secure, standardized, and interoperable framework for using health data across the EU — empowering both citizens and researchers.

Core Objectives

Primary use: Enable individuals to access, control, and share their health data across borders.

Secondary use: Facilitate research, innovation, and policymaking using anonymized or pseudonymized health data.

Interoperability & trust: Establish common standards, governance, and technical frameworks to ensure data portability and security.

European Health Data Space (EHDS): Principles and Vision

- The European Health Data Space (EHDS) is a cornerstone of the European Health Union and the first common EU data space dedicated to a specific sector as part of the European strategy for data
- Aim: **establish a common framework for the use and exchange of electronic health data across the EU**
- EHDS enhances individuals' **access** to and **control** over their personal electronic health data, while also enabling certain **data to be reused** for public interest, policy support, and scientific research purposes.
- It fosters a health-specific data environment that supports a single market for digital health services and products.
- The regulation establishes a **harmonised legal and technical framework** for electronic health record (EHR) systems, fostering interoperability, innovation, and the smooth functioning of the internal market.

https://health.ec.europa.eu/ehealth-digital-health-and-care/european-health-data-space-regulation-ehds_en

European Health Data Space (EHDS)

- The EHDS will:
  - empower individuals to access, control and share their electronic health data across borders for the healthcare delivery (**primary use of data**);
  - enable the secure and trustworthy reuse health data for research, innovation, policy-making, and regulatory activities (**secondary use of data**);
  - foster a single market for electronic health record (EHR) systems, supporting both primary and secondary use.

March 2025

The EHDS Regulation enters into force, marking the beginning of the transition period.
https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ:L_202500327

March 2027

Deadline for the Commission to adopt several key implementing acts, providing detailed rules for the regulation operationalisation

March 2029

Key parts of the EHDS Regulation will enter into application, including, for primary use, the exchange of the first group of priority categories of health data (Patient Summaries, ePrescriptions/eDispensations) in all EU Member States. Rules on secondary use will also start to apply for most data categories (e.g. data from electronic health records).

March 2031

For primary use, the exchange of the second group of priority categories of health data (medical images, lab results, and hospital discharge reports) should be operational in all EU Member States. Rules on secondary use will also start to apply for the remaining data categories (e.g. genomic data).

March 2035

Third countries and international organisations will be able to apply to join HealthData@EU, for the secondary use

Benefits from EHDS

Patients

- fast and free access over their own electronic health data
- easy sharing of health data with health professionals, including across borders
- more control over their own electronic health data: possibility to add personal health information, restrict access to specific parts of their record or to specific persons, view who accessed their data, ask for corrections if errors are found and the right to view health data in a standard European format. (EEHRxF)
- security and privacy protections by default
- opt out rights from secondary use of their own electronic health data.

Health professionals

- faster and easier access to patient's health records across different healthcare providers and countries
- easier access to health records from different systems, significantly reducing the administrative burden.

Benefits from EHDS

Researchers

- access to large-scale health data for scientific
- a clear and structured system to discover what data is available, where it is located, and its quality
- more cost-efficient access to high-quality health data.

Regulators and policymakers

- easier, more transparent and cost-effective access to electronic health data for the public health monitoring, improved healthcare systems efficiency and ensured patient safety.

GDPR



Introduction to GDPR

- EU-wide data protection law implemented in May 2018
- Protects the privacy and rights of individuals regarding their personal data
- Applies to all organizations processing data of people in the EU
- Especially important in healthcare and medical research

Objectives of GDPR

- GDPR aims to:
  - Strengthen data privacy across the EU
  - Give individuals control over their personal data
  - Increase transparency in data processing
  - Ensure that organizations follow responsible data handling practices
  - Support ethically conducted research while protecting participants

What Counts as Personal Data?

- GDPR covers any information that can **identify** a person directly or indirectly.
- Examples include:
  - Name, ID, email, phone number
  - Photos, videos, voice recordings
  - Location data, IP address
  - Biometric identifiers (face, fingerprints)
  - Health data (medical history, diagnoses, test results)
  - Genetic data
  - Pseudonymized data (still considered personal data if re-identification is possible)

GDPR and Medical Research

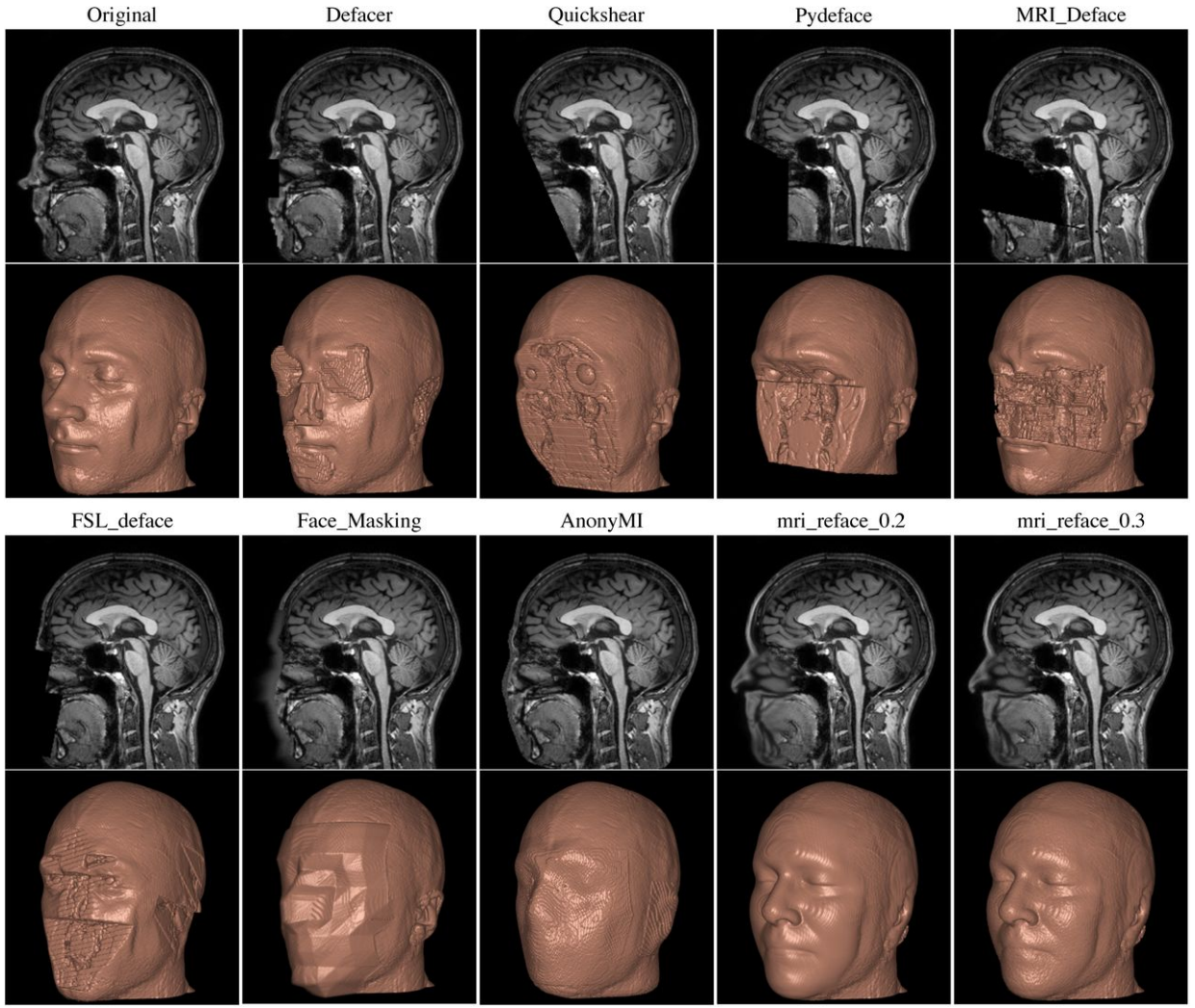
- GDPR pays special attention to **sensitive data**, including:
  - Health information
  - Genetic data
  - Biometric data
  - Racial/ethnic origin
  - Sexual orientation
  - Psychological data
  - Data about children
- GDPR permits the use of personal data for research **under strict conditions**:
  - Must have a **lawful basis** (e.g., public interest, scientific research)
  - Applies **data minimization** (collect only what is necessary)
  - Requires **appropriate safeguards** (pseudonymization, restricted access)
  - Allows **long-term storage** if required for research
  - Informed consent is often used but **not mandatory** if another lawful basis applies

Anonymization vs Pseudonymization

- **Anonymized Data**
  - Cannot identify a person
  - **GDPR does not apply**
  - Often used for data sharing and open datasets
  - Very difficult to guarantee in imaging (e.g., facial reconstruction)
- **Pseudonymized Data**
  - Identifiers replaced by codes
  - **Still covered by GDPR**
  - Widely used in clinical trials and registries

Examples of GDPR in Medical Research

- **Example 1 — MRI Research Study**
  - Facial features must be removed or blurred
  - DICOM tags like name, DOB, ID must be stripped
  - Data stored on secure servers with access control
- **Example 2 — Genomic Dataset**
  - Requires explicit consent or public-interest justification
  - Cannot be fully anonymized → treated as personal data
- **Example 3 — Wearable Device Study**
  - Continuous data tracking → must inform participants
  - Clear retention period defined
  - Right to withdraw must be respected



The Cancer Clinical problems and Challenges

Ioanna Chouvarda, ioannach@auth.gr
Lab of Computing, Medical Informatics and Biomedical-Imaging Technologies, AUTH, GR

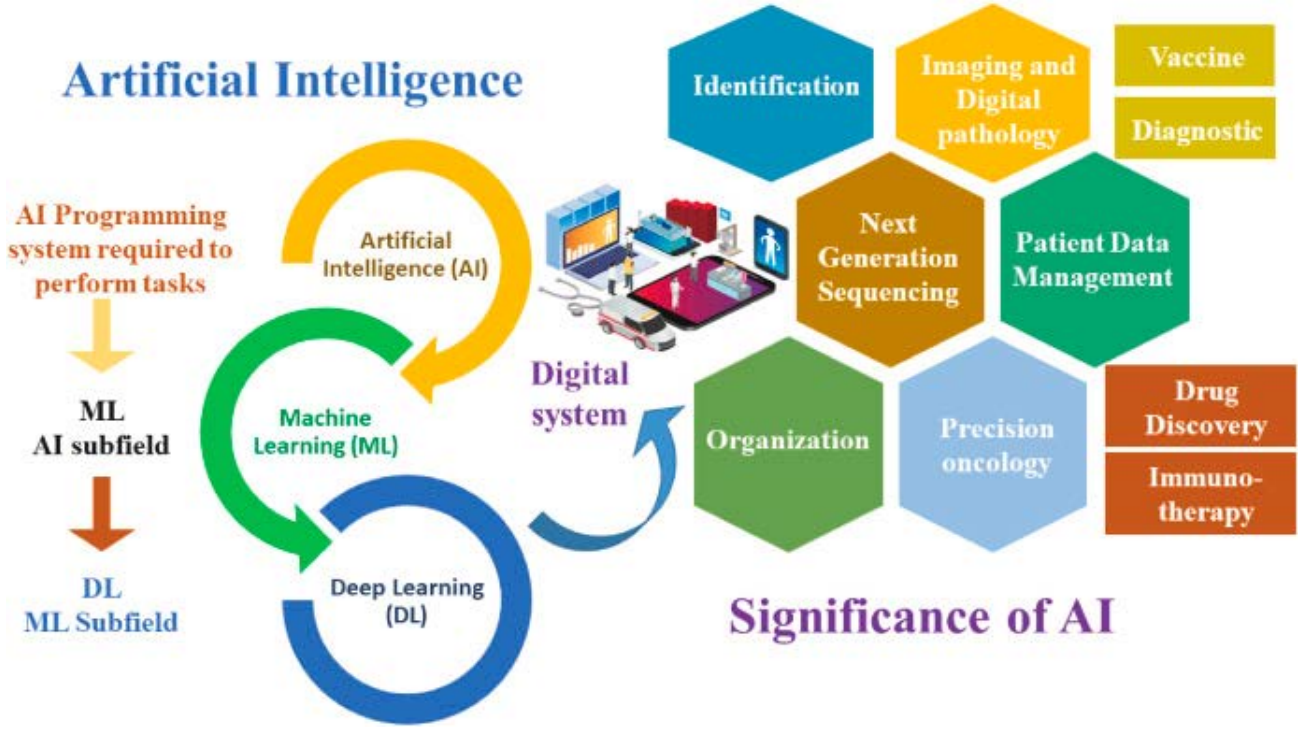
European focus on Cancer



https://research-and-innovation.ec.europa.eu/funding/funding-opportunities/funding-programmes-and-open-calls/horizon-europe/eu-missions-horizon-europe/eu-mission-cancer_en

AI in Medicine

- In data-driven research - Precision
  - Data mining, the ability to uncover hidden ideas and patterns in an independent & unbiased way. Supporting the provision of additional knowledge to health professionals.



- In clinical practice - AI-DSS
  - Automation - Speed - Efficiency - timely information. Solutions that automate time-consuming data processing processes for diagnostic support, e.g. in radiology, ophthalmology, especially when dealing with large volumes and heterogeneous/varied data
  - Accuracy and repeatability. Analysis of multiple data for a more accurate and more objective result

For the Development of Methods

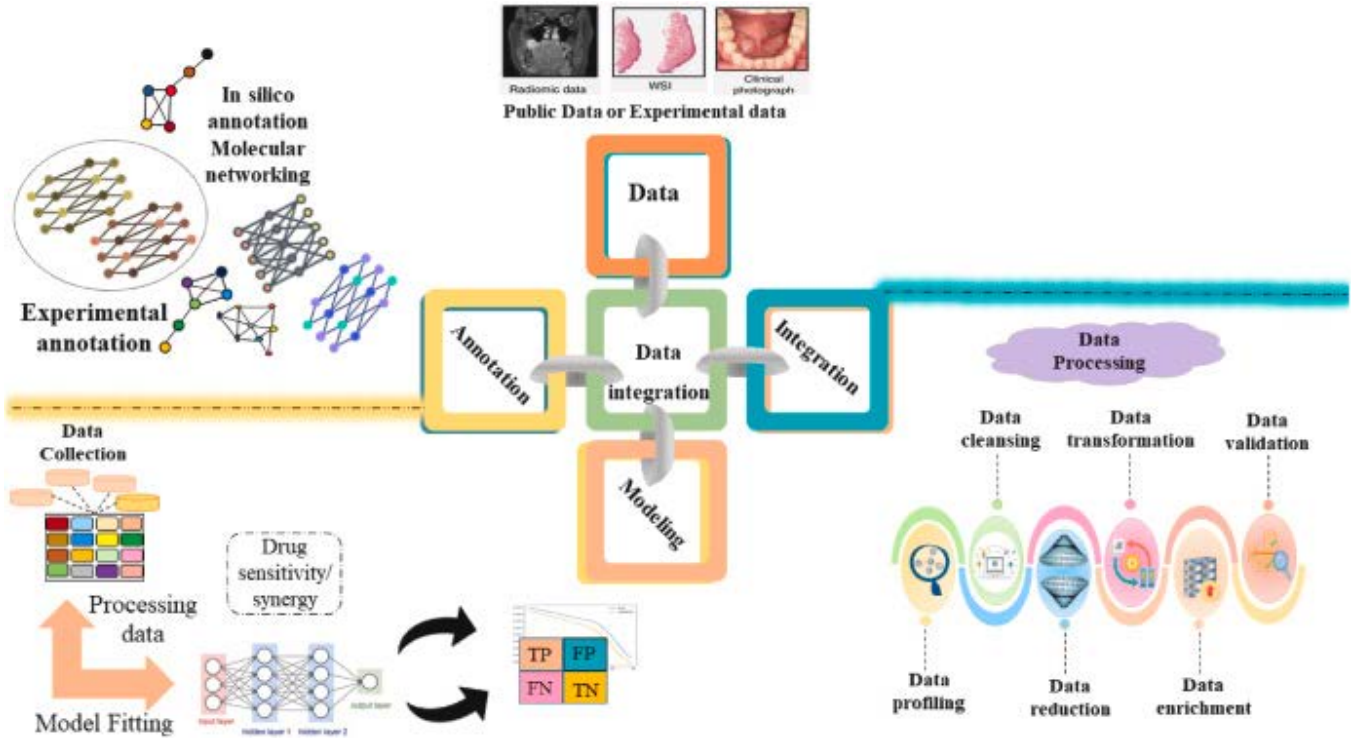
Identifying clinical challenges and mapping them to artificial intelligence models

Data Collection and Data Management, Annotation, Curation

Data processing / computing resources

Domain knowledge + interdisciplinary teams

Ethical and legal issues

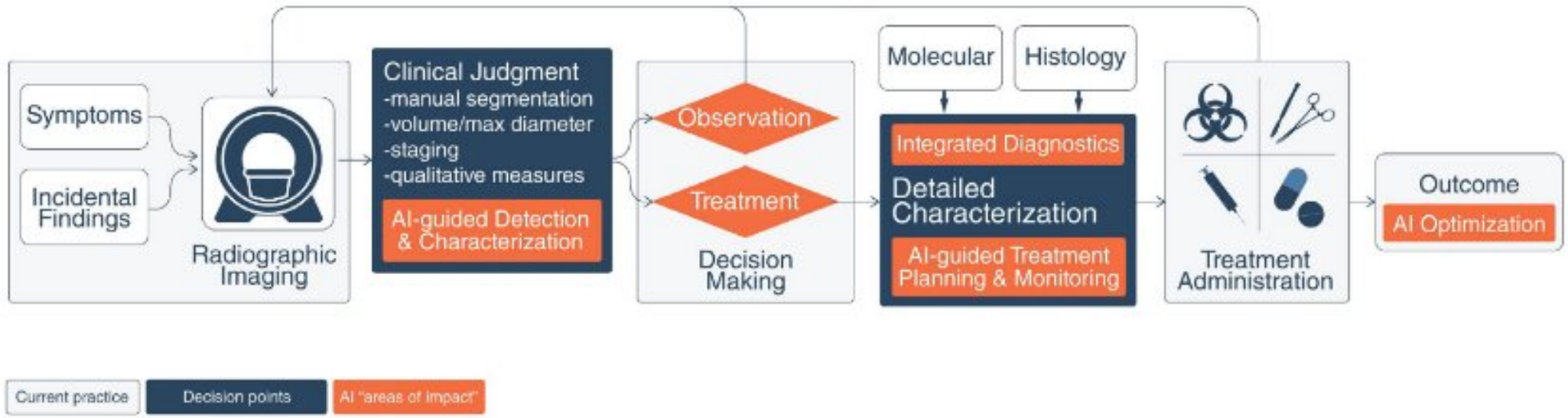


Muhammad Sufyan, Zeeshan Shokat, Usman Ali Ashfaq,Artificial intelligence in cancer diagnosis and therapy: Current status and future perspective,Computers in Biology and Medicine, Volume 165, 2023, 107356, ISSN 0010-4825,https://doi.org/10.1016/j.compbiomed.2023.107356.

Challenges in clinical routine and AI-assisted procedures

What are the clinical pathways - how common are they (eg at EU level)? What points need support? Because of delays? due to reduced accuracy; Where can an AI-based tool be introduced? how will it fit into the process? what is the performance needed?

Lung Cancer: a) segmentation - fast / accurate, b) staging - fast / accurate, c) prediction of metastasis - accurate



Bi, Wenya Linda, et al. "Artificial intelligence in cancer imaging: clinical challenges and applications." *CA: a cancer journal for clinicians* 69.2 (2019): 127-157.

2.1 Lung Cancer

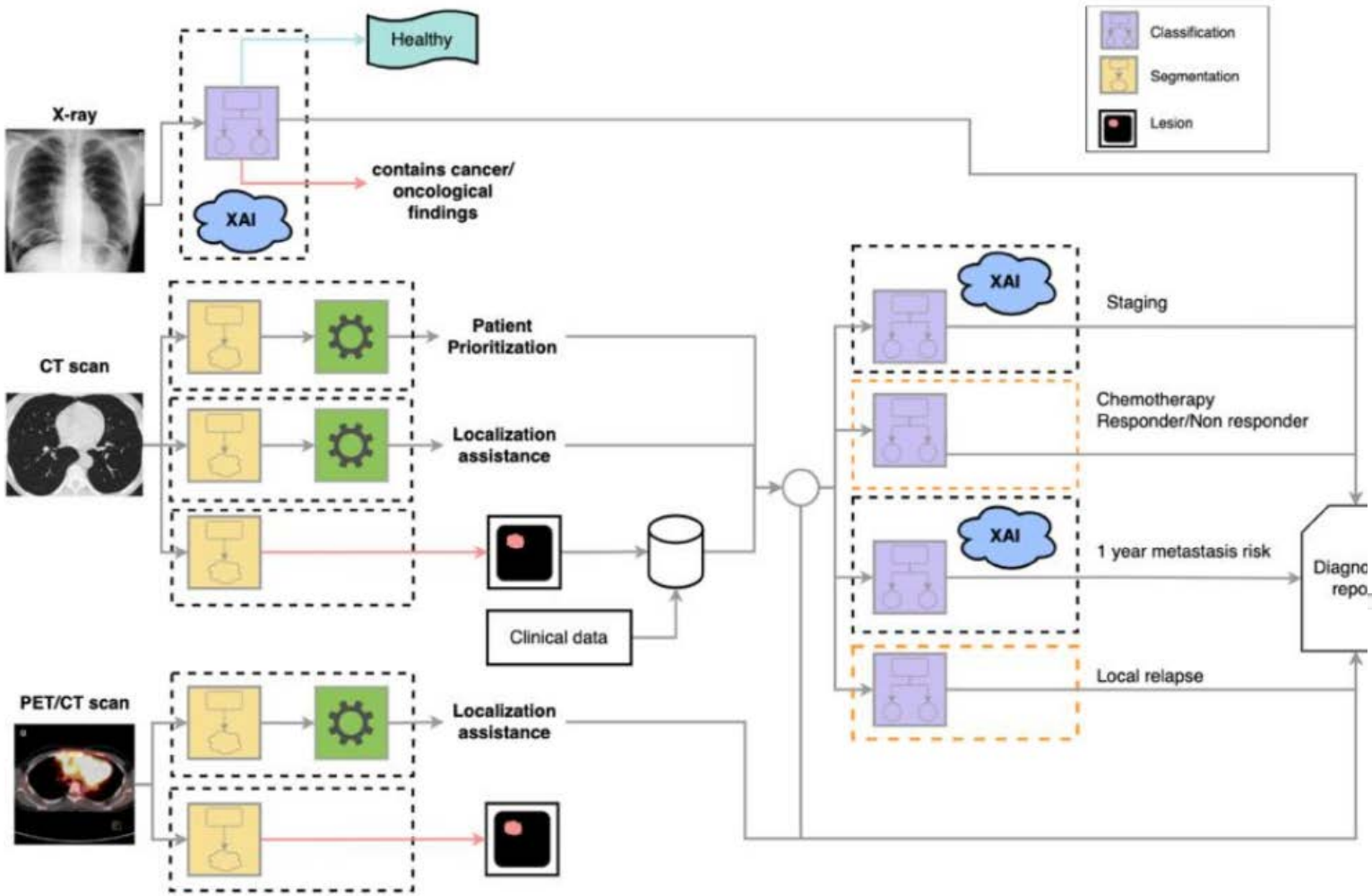


Figure 1: INCISIVE AI services for lung cancer

	Necessary Features of the AI Toolbox	Nice to have Features of the AI toolbox
1.	Classification of tumor as malignant/benign	Assessing intratumor heterogeneity through medical imaging (both through morphological and functional data)
2.	Accurate segmentation of malignant tumor	Cancer staging
3.	Extraction of novel biomarkers/quantitative radiomic features to predict disease severity, invasiveness and progression	Solitary lung nodules detection and classification
4.	Modelling/mapping the potential changes of the tumor in time	Prediction of PD-L1 status, mutations status

5.	Optimization of decision making (risk stratification)	Association of genetic features with imaging
6.	Prediction of metastasis risk	Tumor grading
7.	Prediction/Evaluation of treatment response	Capture discriminative features that go beyond the traditional employed criteria (position, morphology, solid or not)

2.2 Breast Cancer

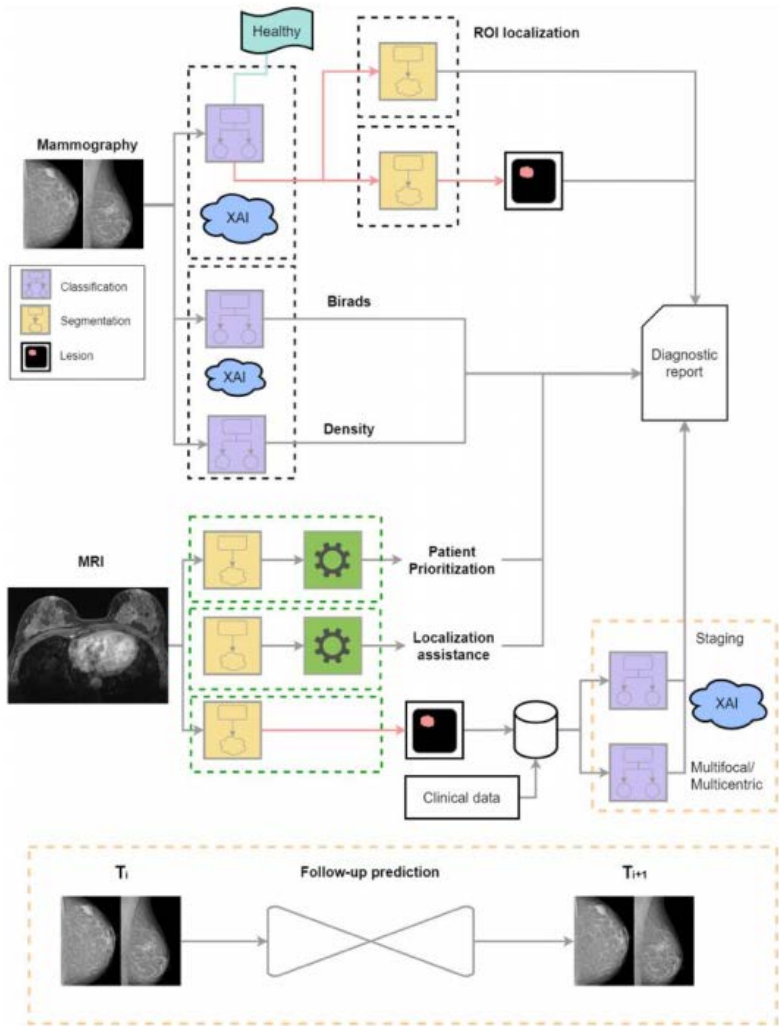


Figure 2: INCISIVE AI services for breast cancer

2.3 Prostate Cancer

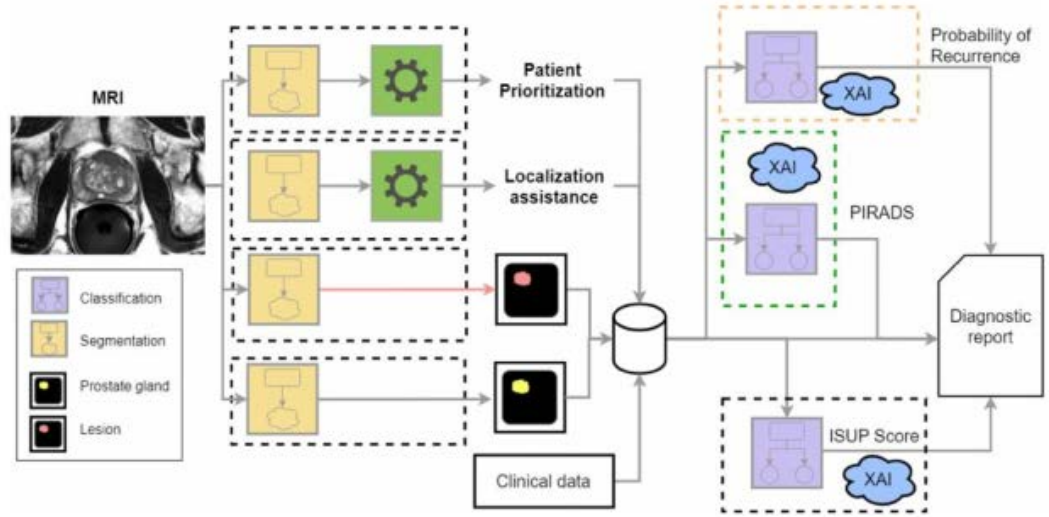


Figure 3: INCISIVE AI services for prostate cancer

2.4 Colorectal Cancer

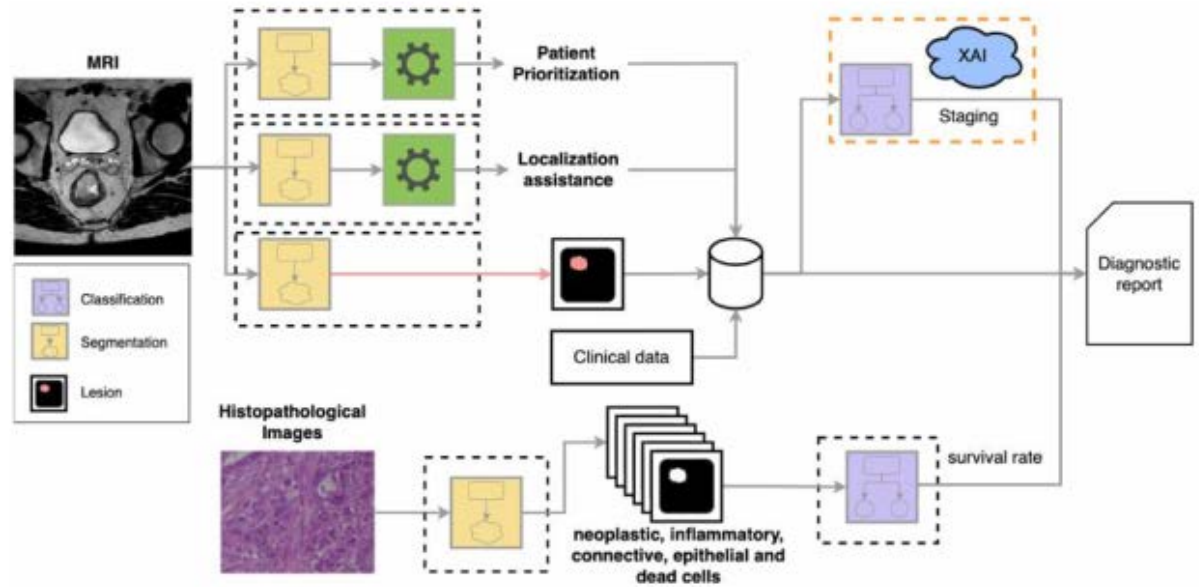


Figure 4: INCISIVE AI services for colorectal cancer

Europe’s Beating Cancer Plan

- **Europe’s Beating Cancer Plan** is a political commitment to turn the tide against cancer and another stepping stone towards a **strong European Health Union** and a more secure, better-prepared and more resilient EU.
- In 2020, **2.7 million people in the European Union were diagnosed with cancer,** and another **1.3 million people lost their lives** to it, including over 2,000 young people. Unless we take decisive action now, cancer cases are set to increase by 24% by 2035, making it the leading cause of death in the EU.
- Europe’s Beating Cancer Plan will have **€4 billion** of funding, including €1.25 billion from the future EU4Health programme.

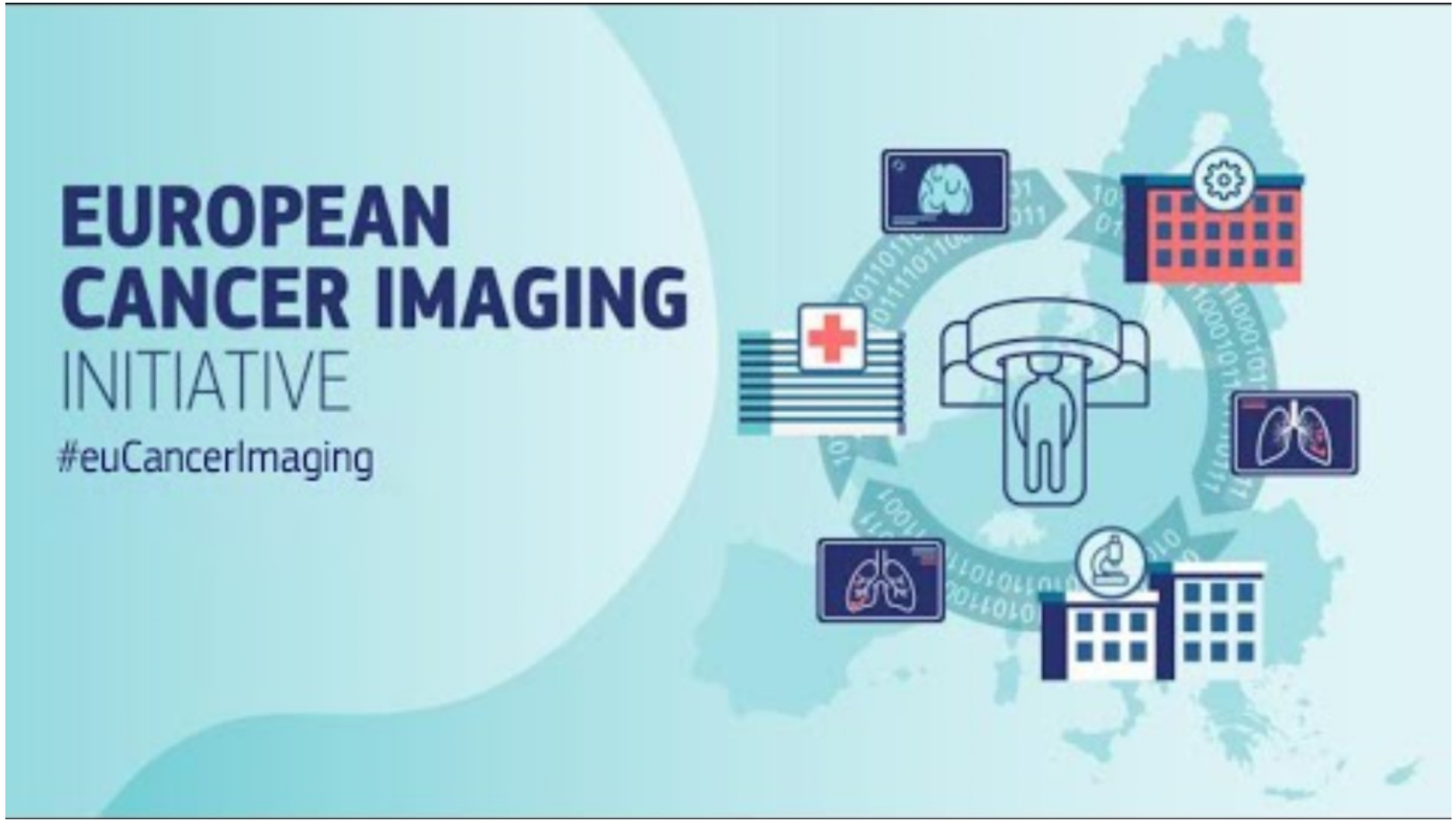
https://commission.europa.eu/topics/public-health/european-health-union/cancer-plan-europe_en

European Cancer Imaging Initiative

- The **European Cancer Imaging Initiative**, one of the flagship actions of Europe's Beating Cancer Plan, aims to harness data, artificial intelligence (AI) and supercomputing to combat cancer more effectively.
- By connecting and sharing imaging and clinical data across Europe, it also supports the European data strategy and the goals of the European Health Data Space.

https://digital-strategy.ec.europa.eu/en/policies/cancer-imaging

The new European Cancer Imaging Initiative



https://www.youtube.com/watch?v=xeFj-8inRAw

Key aims

- Unite and share imaging data - Connect scattered cancer imaging datasets into a secure, open and user-friendly European infrastructure.
- Enable AI-driven innovation - Support the development, benchmarking and clinical validation of AI tools for personalised medicine.
- Foster collaboration - Bring together healthcare providers, researchers and innovators across countries to advance multi-centre studies while respecting data protection rules.
- Ensure trust and ethics - Guarantee the highest standards of data security, patient privacy, transparency and compliance with EU regulations.
- Translate research into care - Harness imaging and clinical data to develop AI solutions that are ready for clinical deployment, improving the processes of early detection, diagnosis and treatment planning

INCISIVE project

Gianna Tsakou

Senior PM/Senior Analyst

INCISIVE Coordinator

INCISIVE

Improving cancer diagnosis and prediction with AI and big data

Key Facts

Title

A multimodal **AI-based toolbox** and an **interoperable health imaging repository** for the empowerment of imaging analysis related to the **diagnosis**, **prediction** and **follow-up** of **cancer**



Duration: 42 months

From October 2020 to March 2024

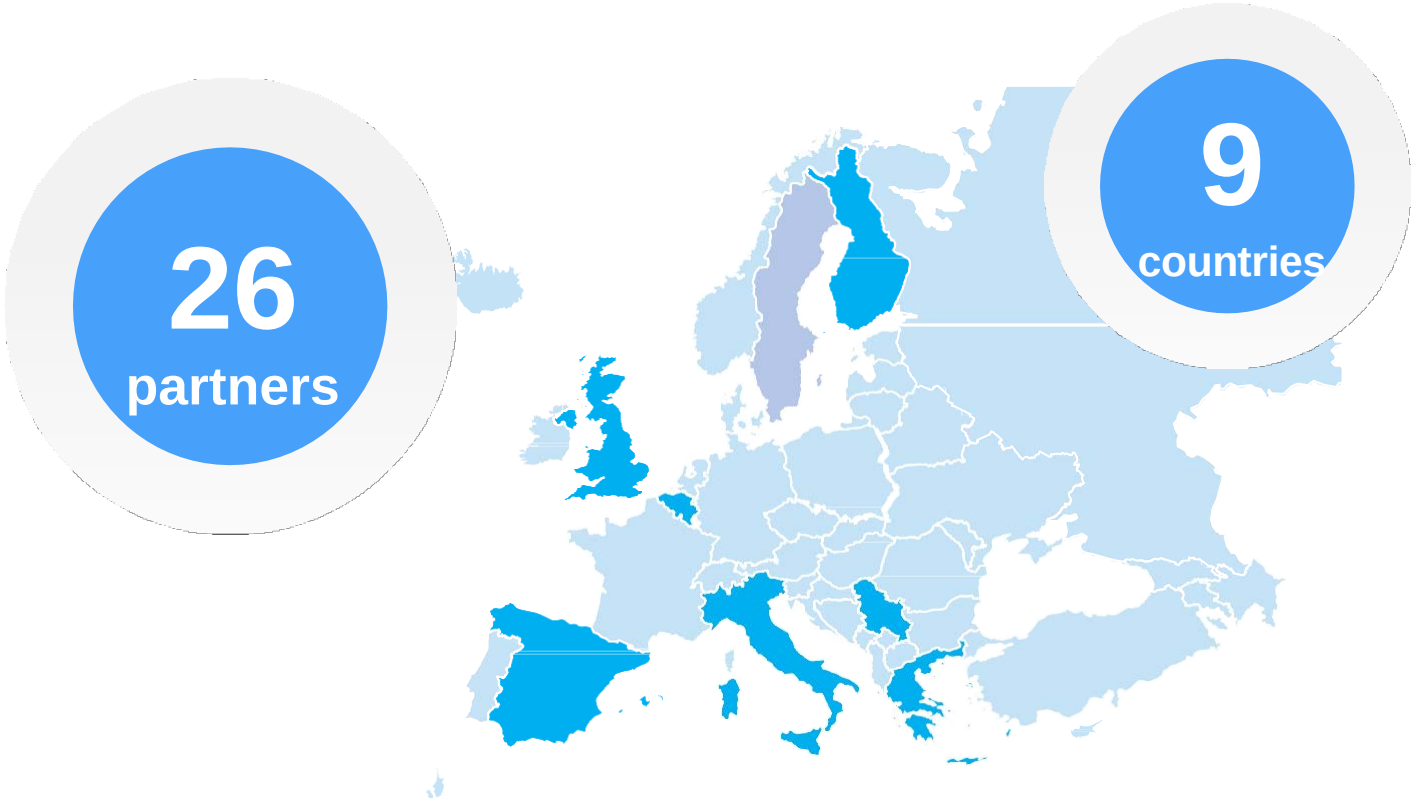
Call: H2020-SC1-FA-DTS-2019

Topic: DT-TDS-05-2020 AI for Health Imaging

Funding: 9.995.727,50 €

This project has received funding from the European's Horizon 2020 research and innovation programme under Grant Agreement number: 952179

Our Consortium



- **9** Universities / Data providers
- **7** SMEs
- **6** Research / academic institutions
- **3** Large industries
- **2** Other

This project has received funding from the European's Horizon 2020 research
and innovation programme under Grant Agreement number: 952179



https://www.youtube.com/watch?v=XFCvxEBOUyg

Areas of expertise

Project Coordination

AI for health imaging & data analytics

Security

High Performance Computing

Complex ICT systems/integration

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Areas of expertise

Cancer clinical research/practice & health data sharing

German Oncology Center

Università di Roma Tor Vergata

IDIBAPS

ΕΛΛΗΝΙΚΗ ΑΝΤΙΚΑΡΚΙΝΙΚΗ ΕΤΑΙΡΕΙΑ
HELLENIC CANCER SOCIETY

VISARIS

Patient representation & evaluation

Cyprus University of Technology

PASYKAF

Kingston University London

Legal and ethical issues

TIMELEX

Innovation and business planning

WHITE RESEARCH

Medtronic
Further, Together

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Main objectives



AI-based toolbox that supports the decision-making of HCPs in relation to patient prioritization, cancer diagnosis and risk for metastasis



Interoperable pan-European **federated health data repository** (medical images & clinical data) that enables secure data sharing in compliance with ethical, legal and privacy requirements

TARGETED IMPACT ON:

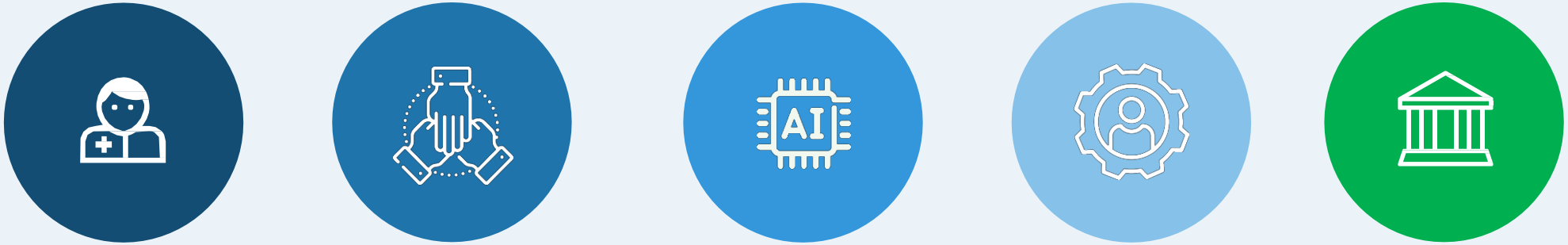
- Healthcare professionals involved in cancer care
- AI developers
  AI experts
- Researchers

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User-driven Approach

NUMEROUS CONSULTATION ACTIVITIES WITH END-USERS

UX WORKSHOPS, SURVEYS, INTERVIEWS, LITERATURE REVIEW, …



Healthcare professionals

Patients' associations

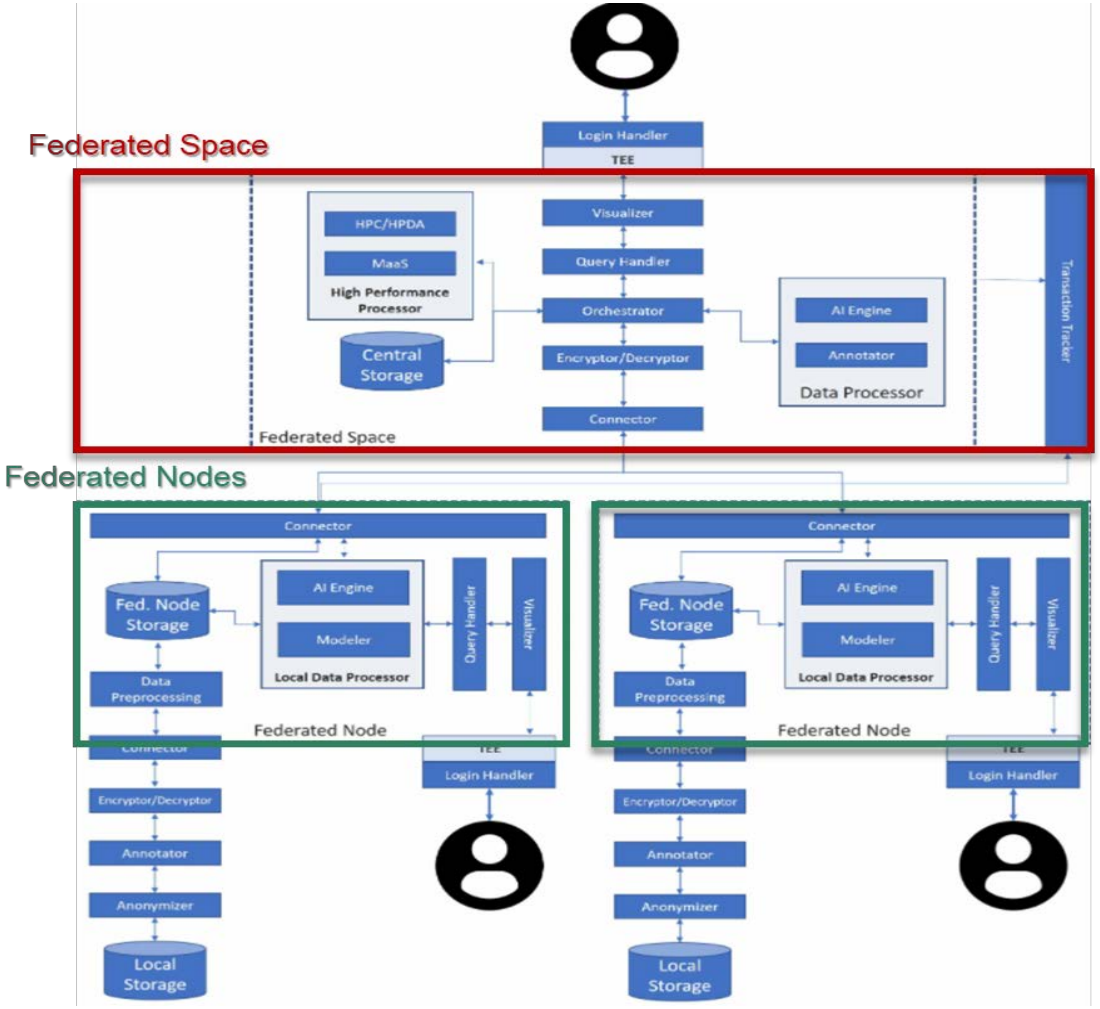
AI researchers

Technology developers

Legal and ethics experts

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and innovation programme under Grant Agreement number: 952179

Data-driven & Federated Approach



- **Hybrid storage:**
  - **Both federated and central nodes available**
  - **Selection based on data provider's needs**
  - **Federated Learning mechanisms**

Data federation facilitates:

- aggregation of data from multiple sources
- ethical, legal and privacy compliance

This project has received funding from the European's Horizon 2020 research and innovation programme under Grant Agreement number: 952179

Clinical & Tech pillars

4 cycles

Data collection

Retrospective data

- for AI training

Prospective data

- for AI training
- AI validation – observational study
- AI feasibility testing

AI Development

AI models' development in three iterations

AI toobox prototypes

First M18

second M30

final M42

Tech Infrastructure

Supports development and delivery of all services

Centralised space

•central data storage

Federated space

- distributed data storage

Federated Learning

- Distributed data processing tools

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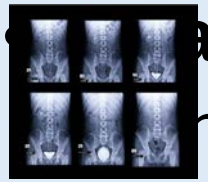
Obj. 1 achievements: AI toolbox

XAI, Federated Learning
Models-as-a-Service



- **Lung cancer:** classification of abnormalities, patient prioritization, localization assistance, cancer staging, 1-year metastasis risk prediction *Modalities:* X-ray, CT, PET-CT





- **Breast cancer:** classification of abnormalities, lesion segmentation, patient prioritization, localization assistance, breast density classification, BIRADS classification *Modalities:* mammography, MRI



- **Prostate cancer:** instance segmentation for patient prioritization, lesion segmentation for localization assistance, BIRADS score prediction, Gleason score prediction

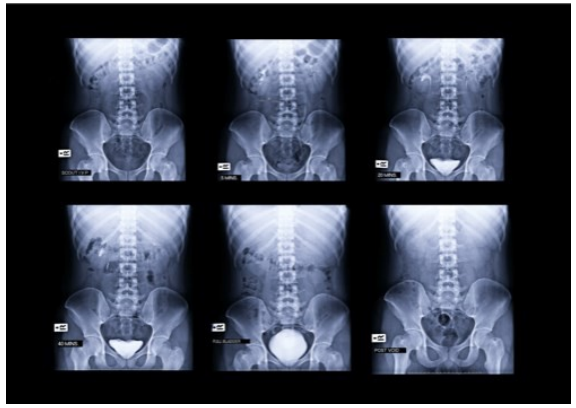
This project has received funding from the European's Horizon 2020 research and innovation programme under Grant Agreement number: 952179

Obj. 1 achievements: AI toolbox - Pilot studies



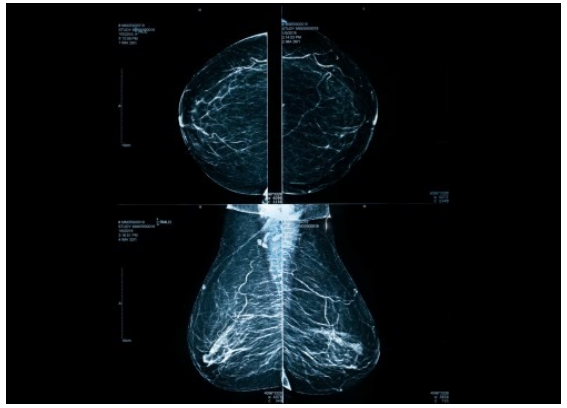
LUNG CANCER

(Cyprus, Greece, Italy)



PROSTATE CANCER

(Cyprus, Greece, Spain)



BREAST CANCER

(Cyprus, Greece, Italy, Serbia)



COLORECTAL CANCER

(Cyprus, Greece, Italy)

2,000 patients

participated in 8 pilot studies in 5 countries

This project has received funding from the European's Horizon 2020 research and innovation programme under Grant Agreement number: 952179

Obj. 2 achievements: Data collection

- **Imaging** & accompanying **clinical & biological data**
- **Secure, transparent, GDPR-compliant** data sharing according to **FAIR** principles (**F**indability**, A**ccessibility**, I**nteroperability and **R**eusability)

- **9 Data Providers from 5 countries**
  - 3 hospitals as consortium partners
  - 4 Universities collecting data from 6 associated hospitals
  - 1 NGO collecting data from 2 associated hospitals
  - 1 SME (X-ray equipment manufacturer)



This project has received funding from the European's Horizon 2020 research and innovation programme under Grant Agreement number: 952179

Obj. 2 achievements: Data Types

- **Imaging Data (at diagnosis and subsequent timepoints)**
  - MRI
  - CT
  - Ultrasound
  - PET-CT
  - Mammography
  - X-ray
  - Histopathological images
- **Accompanying Clinical Data (at diagnosis and timepoints)**
  - Demographics, treatment, mutations, lab results
  - Excel Template used for collection



This project has received funding from the European's Horizon 2020 research and innovation programme under Grant Agreement number: 952179

Obj. 2 achievements: Data collection

Steps followed

ETHICAL APPROVAL ACTIONS

- From M3-M12

PATIENT SELECTION

- Medical image type and quality specifications
- Within hospital data collection
- Complete cases – Incomplete cases (less time points available)

IMAGE DE-IDENTIFICATION

- Comomn de-identification protocol defined
- CTP anonymized →
- INCISIVE-specific de-identification tool

CLINICAL DATA EXTRACTION

- Extraction of clinical data from HIS into predefined data templates

DATA QUALITY CHECK

- checks various data quality and structuring parameters

IMAGE ANNOTATION

- Annotation guidelines
- ITK SNAP tool →
- INCISIVE-specific annotation tool

DATA HOMOGENISATION

- Mapping to medical ontologies SNOMED-CT & LOINC (semantic)
- Transformation to FHIR data model (structural)

DATA UPLOAD to INCISIVE REPOSITORY

- ETL tool for interoperable data upload

This project has received funding from the European's Horizon 2020 research and innovation programme under Grant Agreement number: 952179

Data Sharing Framework

INCISIVE data infrastructure

meets

LEGAL/ETHICAL REQUIREMENTS

- Data governance and access management framework
- GDRP compliance
- Support for de-identification & annotation of health images, quality check of clinical data – tools and guidance

RESEARCHERS NEEDS

- Making the data available for re-use by researchers, incl. AI developers
- Multi-centre, multi-country data sources
- Standardised, interoperable, quality-checked data

facilitates

data holders to share medical images & health data with minimum effort

OPEN INVITATION to interested Data Providers!

This project has received funding from the European's Horizon 2020 research and innovation programme under Grant Agreement number: 952179

INCISIVE's Legacy

AI toolbox: AI models & pipelines, incl. FL mechanism

28 AI models/services implemented

4 types of cancer

Hybrid Data Repository infrastructure

>5,5M cancer images

>11,000 patients

Data interoperability framework (methodology & tools, incl. CDM)

5 federated data nodes + 1 central fully operational

9/9 data providers integrated / federated repository

This project has received funding from the European's Horizon 2020 research and innovation programme under Grant Agreement number: 952179

INCISIVE Sustainability



https://www.youtube.com/@EUCAIM

This project has received funding from the European's Horizon 2020 research and innovation programme under Grant Agreement number: 952179


CANCER IMAGE EUROPE
AI

EDIC Roadmap: participant countries

Countries having already appointed ministerial representatives via LOI:



https://www.youtube.com/watch?v=Kt1wlp6ByKw

https://www.youtube.com/watch?v=7HbM-Yjjs7I

EUCAIM overview

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AUTH

The background: AI4HI cluster

Project	Goal	Considered use cases	Types of metadata	Adopted models
PRIMAGE	To build an imaging biobank for the training and validation of machine learning and multiscale simulation algorithms	Paediatric neuroblastoma and diffuse intrinsic pontine glioma	DICOM tags Image analysis metadata (registration, denoising, radiomics) Clinical variables	DICOM-MIABIS OMOP CDM
EuCanImage	To build a European cancer imaging platform for enhanced AI in oncology	Eight use cases regarding liver, breast, and colorectal cancer	Imaging data Clinical variables	DICOM-MIABIS ICGC-ARGO
INCISIVE	To improve cancer diagnosis and prediction with AI and big data	Lung, breast, colorectal, and prostate cancer	Imaging data Clinical and biological data	FHIR
CHAIMELEON	To develop a structured repository of health images and related clinical and molecular data	Lung, breast, prostate, and colorectal cancer	Imaging data Clinical variables	DICOM-MIABIS OMOP CDM
ProCancer-I	To develop an AI Platform integrating imaging data and models	Prostate cancer	Imaging data Clinical variables	DICOM-Radiation therapy OMOP CDM with Oncology Extension

Kondylakis H, Ciarrocchi E, Cerda-Alberich L et al (2022) Position of the AI for Health Imaging (AI4HI) network on metadata models for imaging biobanks. Eur Radiol Exp 6:29. https://doi.org/10.1186/s41747-022-00281-1

Kondylakis, H., Ciarrocchi, E., Cerda-Alberich, L. *et al.* Position of the AI for Health Imaging (AI4HI) network on metadata models for imaging biobanks. *Eur Radiol Exp* **6**, 29 (2022). https://doi.org/10.1186/s41747-022-00281-1

Value Proposition

EUCAIM
CANCER IMAGE EUROPE

A pan-European federated infrastructure, linking and making research cancer imaging data from EU-level and national hospital networks and research repositories **accesible**, **ready** & **usable** .

An **accelerator** for medical research and industry to better understand, diagnose and fight cancer.

https://cancerimage.eu/

Target Groups

DATA HOLDERS

They provide access to the cancer imaging and related data to be shared through EUCAIM



- Clinical sites
- Existing cancer image repositories
- Research infrastructures
- Cancer screening programmes
- Others (e.g. data altruism organisations, etc.)

DATA USERS

They use the EUCAIM infrastructure to easy or enrich their achievement of research results



- Researchers & innovators in data science & AI
- Researchers & innovators in new therapies
- Medicines regulatory agencies
- Others (e.g. policymakers, etc.)

EUCAIM interoperability

Interoperable: utilization of common interoperability mechanisms developed and used in the AI4HI projects (CHAIMELEON, EUCANIMAGE,INCISIVE, PRIMAGE, and ProCANCER-I) and the major Research Infrastructures dealing with cancer data (Euro-BioImaging, BBMRI-ERIC, EATRIS, ECRIN and ELIXIR).

- The infrastructure will be interoperable with the European Health Data Space (EHDS) components, the initiative of Technical and Experimentation Facilities (TEF), and the developments of the AI Factories under the European High Performance Computing (EuroHPC) Joint Undertaking.

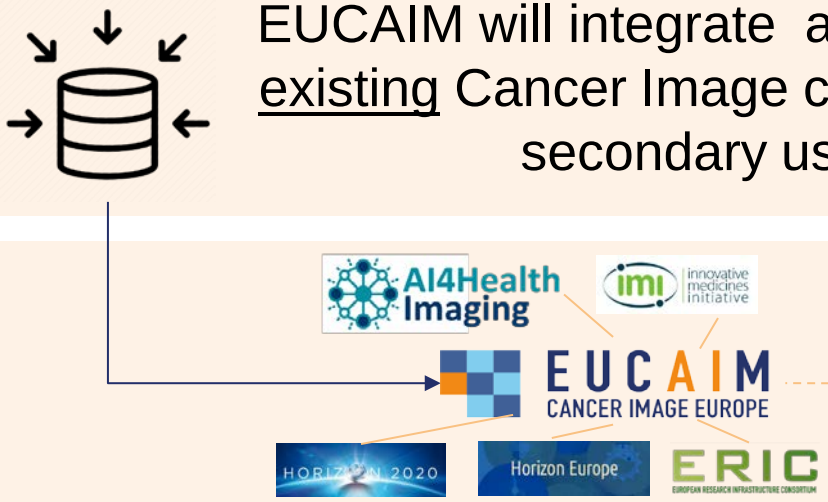
Value Proposition

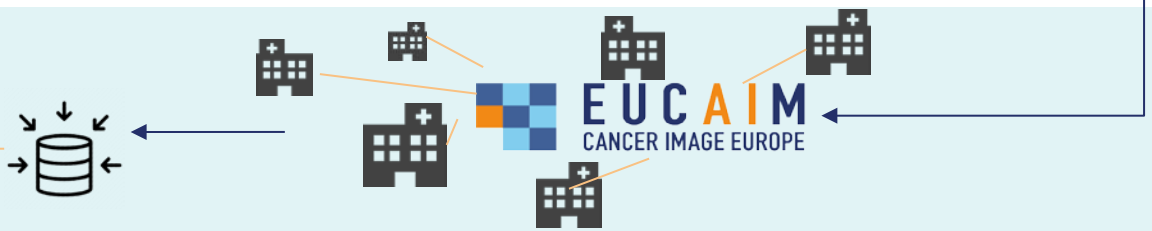
ATLAS OF CANCER IMAGES

EUCAIM will integrate and capitalize existing Cancer Image collections for secondary use

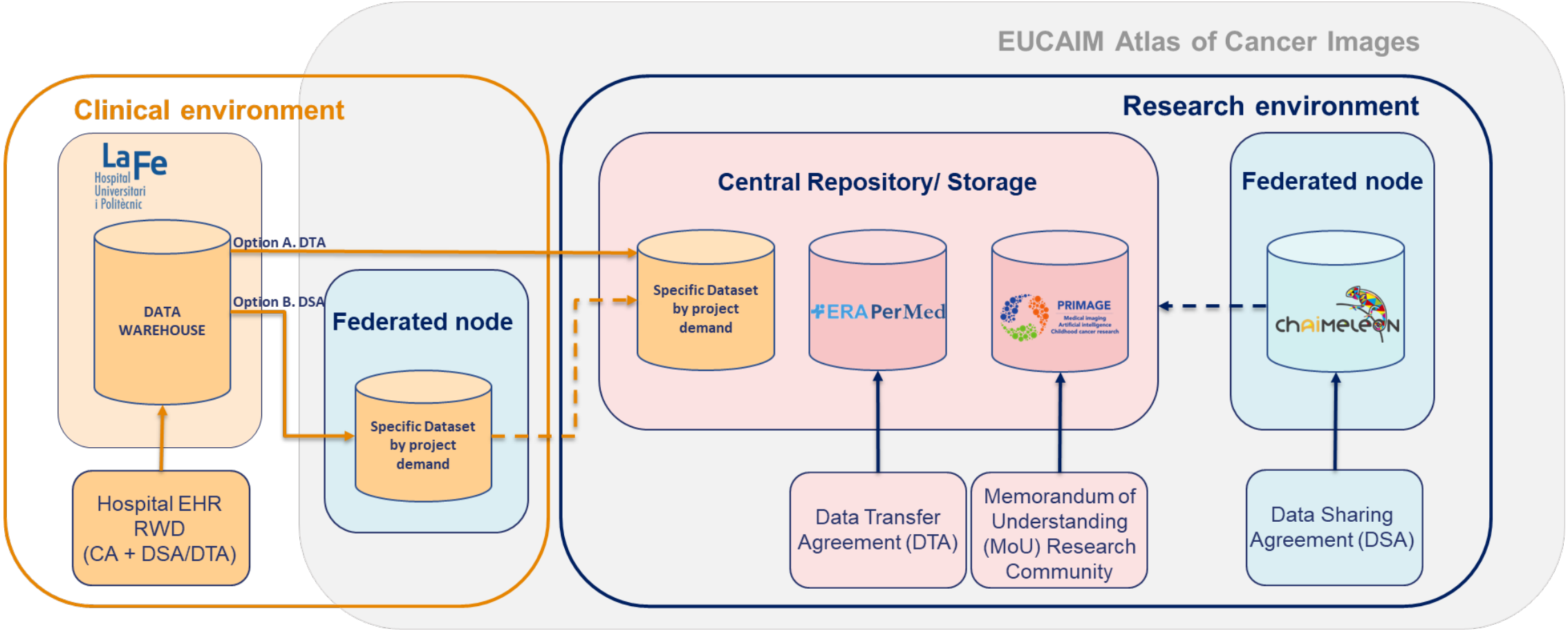
OBSERVATIONAL STUDIES + CANCER SCREENING

EUCAIM will make Cancer Image collections newly available to address specific Research Questions





EUCAIM & THE ATLAS OF CANCER IMAGES
Central Storage vs Federated



TIERs: Data compliance levels in EUCAIM

Support data curation and facilitate data transformation, if necessary, to align with the Data Federation Framework.
Use the Observational Medical Outcomes Partnership (OMOP) to enable the integration of the multiple data sources and to support the analysis and exploitation capabilities of the datasets.
To accommodate diverse levels of data compliance, it has been established three technical tiers. These tiers are scalable, allowing upgrading as the datasets are used in new research projects. EUCAIM will provide support and adaptation plans to assist the achievement of Tier 2 and Tier 3 compliance.

Tier 1

Acceptance of data with no DFF compliance, suitable for entry-level participation.

Interoperability is required at the level of the dataset's aggregated metadata, so the data is findable through the infrastructure's metadata catalogue.

Tier 2

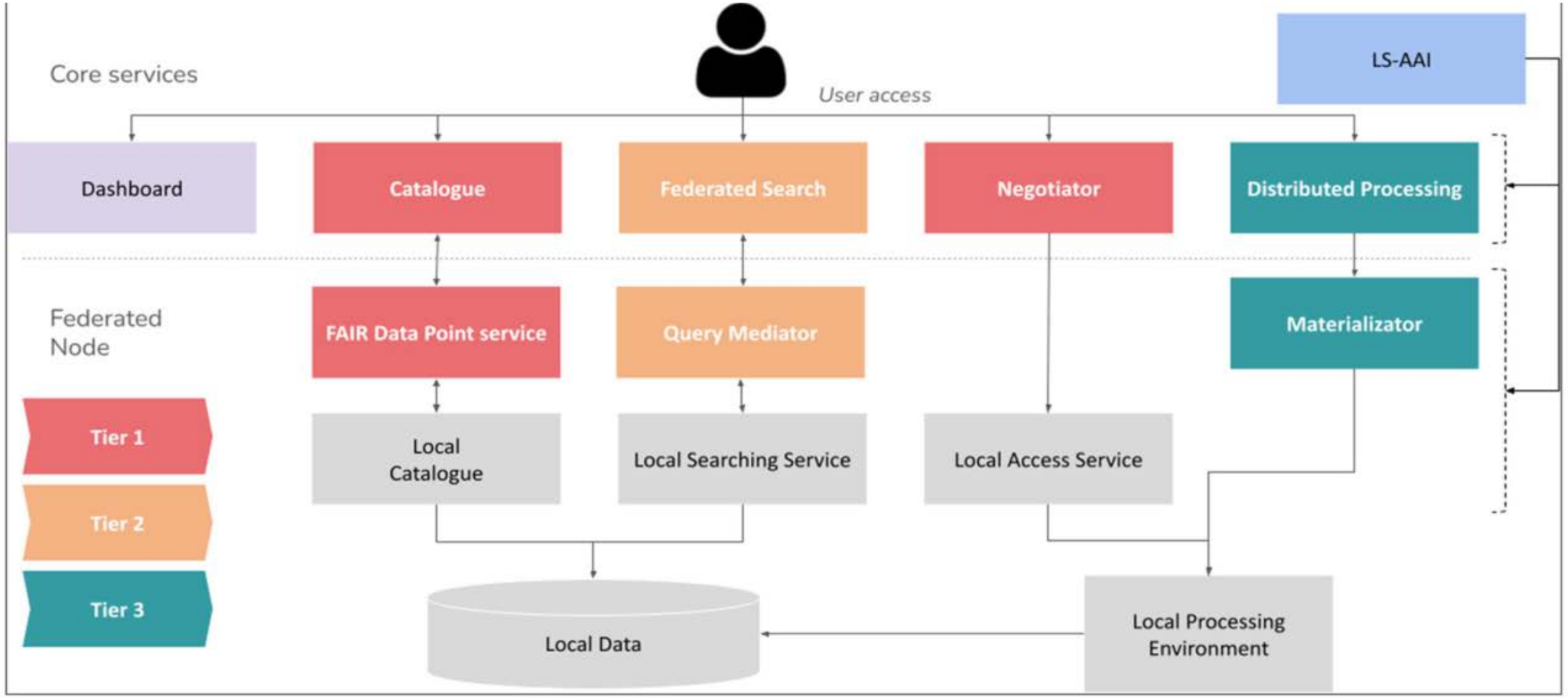
Medium compliance with the DFF, enabling federated search functionality.

Interoperability is also implemented at the level of the federated search, ensuring that the metadata is accessible.

Tier 3

- Full compliance with the DFF, allowing distributed processing, including machine learning model training.
- Interoperability also includes integration into the federated processing service.

Architecture



Martí-Bonmatí, L., Blanquer, I., Tsiknakis, M. *et al.* Empowering cancer research in Europe: the EUCAIM cancer imaging infrastructure. *Insights Imaging* **16**, 47 (2025). https://doi.org/10.1186/s13244-025-01913-x

Definition of common data and metadata models

- EUCAIM has defined a Minimum Common Data Model for the data and metadata, and the protocols, formats, and terminologies for the clinical and imaging data.
- After the thorough comparison of existing common data models, such as OMOP-CDM, and catalogue specifications (e.g., DCAT-AP for Health and MIABIS), EUCAIM developed a first version of a hyperontology (or hyper-data model) that enables the interoperability of all AI4H repositories and with any other data collected in the EUCAIM federated repository.

- EUCAIM's HyperOntology_v0.1beta. Zenodo https://doi.org/10.5281/zenodo.10777926
- See also https://confluence.hl7.org/spaces/COD/pages/91980352/mCODE

Data Preprocessing

- Tools and criteria for assessing data quality, both at the level of each data item (incomplete, low-resolution, noisy data) and at the dataset level (low number of cases, repeated cases).
- Data Harmonization Tools for increasing the comparability of different data providers, which employ statistical and AI-driven methods.
- Data Cleaning tools for detecting and correcting incomplete, incorrect, inaccurate, irrelevant parts of the data and then replacing, modifying, or deleting dirty data.
- Pseudonymization and/or Anonymization approaches have been defined and agreed upon among the data holders, ensuring compliance with the different regulations.
- Data Annotation Tools and Services, already available by consortium members, leveraged and integrated into the EUCAIM infrastructure.

Data quality



https://www.youtube.com/watch?v=J0XTjqNpc-U

Big Repositories

- https://data.mendeley.com/
- Zenodo
- Kaggle
- TCIA
- Grand Challenge
- +++

General / Multimodal Medical Imaging Repositories

- The Cancer Imaging Archive (TCIA)
  - https://www.cancerimagingarchive.net/
- Medical Segmentation Decathlon (MSD)
  - http://medicaldecathlon.com/
  - https://arxiv.org/pdf/1902.09063
- Grand Challenges (grand-challenge.org)
  - https://grand-challenge.org/
- Imaging Data Commons (IDC, part of NCI)
  - https://portal.imaging.datacommons.cancer.gov/
- OpenNeuro
  - https://openneuro.org/
- OASIS (Open Access Series of Imaging Studies)
  - https://sites.wustl.edu/oasisbrains/
- UK Biobank Imaging
  - https://www.ukbiobank.ac.uk/about-our-data/types-of-data/imaging-data/

EchoNet-Dynamic

- **Imaging modality**: Echo 2D
- Gb: 7gb
- **File Format:** The apical-4-chamber view video (AVI) was identified by DICOM file linked to measurements of ventricular volume used to calculate the ejection fraction in the apical-4-chamber view.
- **Open/ Private (costs):** O
- **Number of examples:** 10,030 labeled echo videos and human expert annotations
- **Annotations/ metadata:** -Tracings, Measurements of volumes
- **Region:** 4 chambers
- **Cohort Group:** random individuals who underwent echocardiography between 2006 and 2018 as part of clinical care at a University Hospital.
- **Link:** https://stanfordaimi.azurewebsites.net/datasets/834e1cd1-92f7-4268-9daa-d359198b310a
- **Code or examples of use**: https://echonet.github.io/dynamic/index.html

EchoNet-LVH

- Hypertrophy cardiomyopathy and cardiac amyloidosis.
- **Imaging modality**: 2D echocardiography videos
- **File Format:**
- **Open/ Private (costs):** O (register in Standford AIMI Shared Datasets)
- **Number of examples:** 12,000 labeled echo videos
- **Annotations/ metadata:** human expert annotations (measurements, tracings, and calculations). During both systole and diastole, clinicians measure intraventricular septum (IVS) thickness, left ventricular internal dimension (LVID), and left ventricular posterior wall (LVPW) thickness in the parasternal long axis view.
- **Region:** parasternal long axis view
- **Study population: The EchoNet-LVH dataset contains 12,000 parasternal-long-axis echocardiography videos from individuals who underwent imaging as part of routine clinical care at Stanford Medicine**
- **Link:** https://stanfordaimi.azurewebsites.net/datasets/5b7fcc28-579c-4285-8b72-e4238eac7bd1
- **Code or examples of use:** https://github.com/echonet/lvh

CAMUS

- **Imaging modality**: Echo 2D
- **File Format:** raw/mhd
- **Open/ Private (costs)**: O
- **Number of examples:** 500 patients x 1 cycle
- **Annotations/ metadata:**
- **Region**: four chamber and two-chamber view sequences
- **Study population:** The proposed dataset consists of clinical exams from 500 patients, half of the dataset population has a LVEF lower than 45%, thus being considered at pathological risk (beyond the uncertainty of the measurement). Also, 19% of the images have a poor quality (based on the opinion of one expert O1a), indicating that for this subgroup the localization of the LVEndo and LVEpi as well as the estimation of clinical indices are not considered clinically accurate and workable.
- **Link**: https://www.creatis.insa-lyon.fr/Challenge/camus/scientificInterests.html
- **Code or examples of use:** https://github.com/vitalab/castor

Angiographic dataset for stenosis detection

- **Imaging modality**: invasive coronary angiography
- **File Format:**
- **Open/ Private (costs)**: O
- **Number of examples:** 8325 grayscale images, 100 patients
- **Annotations/ metadata:** plaque labels segmentation
- **Region**: coronary arteries
- **Study population:** All patients had angiographically and/or functionally confirmed one-vessel coronary artery disease (≥70% diameter stenosis by quantitative coronary analysis or 50 - 69% with FFR (fractional flow reserve) ≤ 0.80 or stress echocardiography evidence of regional ischemia).
- **Link**: https://data.mendeley.com/datasets/ydrm75xywg/1

- IXI Dataset
- ADNI (Alzheimer's Disease Neuroimaging Initiative)
  - https://adni.loni.usc.edu/data-samples/adni-data/#AccessData
- **Automated Cardiac Diagnosis Challenge (ACDC)**
  - https://www.creatis.insa-lyon.fr/Challenge/acdc/databases.html

Ultrasound

- POCUS Datasets
  - https://figshare.com/articles/dataset/POCUS_LUS_Datasets/29364743
- OpenFAST Ultrasound Dataset
- Thyroid Ultrasound Cine-clip
  - https://aimi.stanford.edu/datasets/thyroid-ultrasound-cine-clip
- A collection of open access ultrasound imaging data sets.
  - https://ultrasound-open-access.nidusai.ca/

X-ray Repositories

- **NIH Chest X-rays**
- https://www.kaggle.com/datasets/nih-chest-xrays/data
- CheXpert
  - https://stanfordmlgroup.github.io/competitions/chexpert/
- MIMIC-CXR
  - https://physionet.org/content/mimic-cxr/2.1.0/
- VinDr-CXR
  - https://physionet.org/content/vindr-cxr/1.0.0/

Angiography

- ARCADE: Automatic Region-based Coronary Artery Disease diagnostics using x-ray angiography imagEs Dataset
  - https://zenodo.org/records/10390295
- Digital Subtraction Angiography (DSA) / DSA Sequences
  - https://github.com/jiongzhang-john/DSCA?tab=readme-ov-file

Digital Pathology (Whole Slide Imaging)

- PANDA Challenge (Prostate biopsy slides)
  - https://panda.grand-challenge.org/
- PAIP Challenge datasets
  - https://paip2019.grand-challenge.org/Dataset/
- The Cancer Genome Atlas (TCGA) – pathology slides
  - https://cancer.digitalslidearchive.org/#!/CDSA/acc/TCGA-OR-A5J1

Ophthalmology Imaging (Retina)

- EyePACS (Kaggle)
  - https://www.kaggle.com/datasets/ascanipek/eyepacs-aptos-messidor-diabetic-retinopathy
- Messidor Dataset
  - https://www.adcis.net/en/third-party/messidor/
  - https://www.kaggle.com/datasets/mariaherrerot/messidor2preprocess
- **REFUGE2**
  - https://www.kaggle.com/datasets/victorlemosml/refuge2

Dermatology Imaging

- ISIC Archive
  - https://www.isic-archive.com/
- HAM10000
  - https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DBW86T

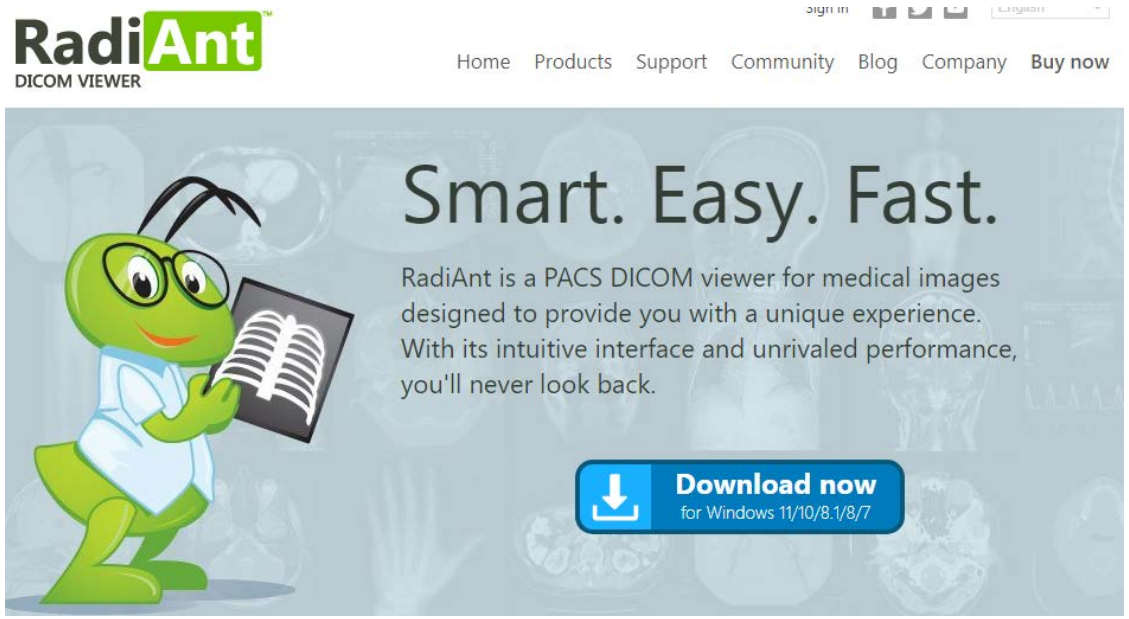
Spinal cord

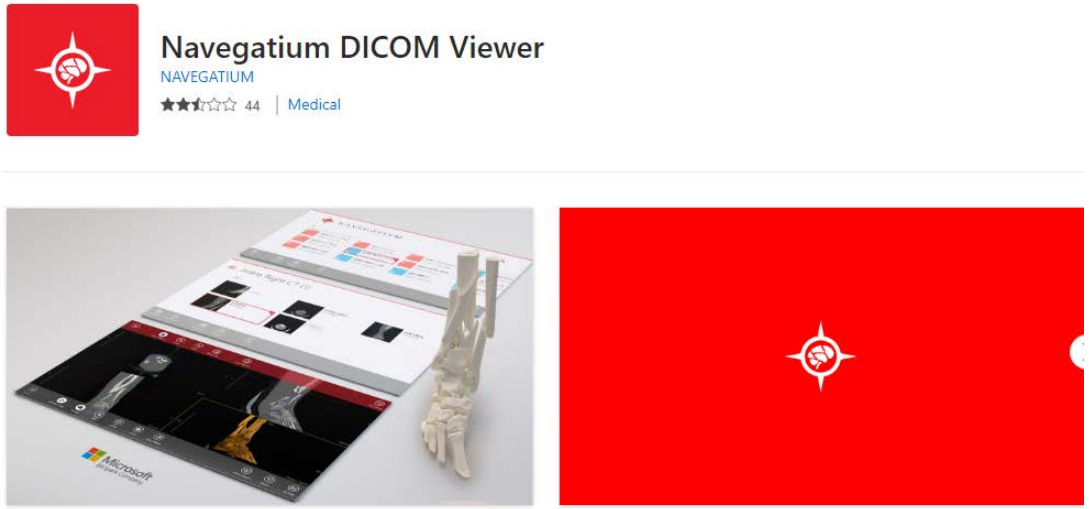
- Open Data Commons for Spinal Cord Injury (ODC-SCI)
  - https://odc-sci.org/
- **Instrumented Spinal Cord T1/T2-weighted MRI Scans**
  - https://data.mendeley.com/datasets/sjpx7md7cf/1

Animals (e.g. Dogs)

- ICDC-GLIOMA (Canine Glioma)
  - https://www.cancerimagingarchive.net/collection/icdc-glioma/

Dicom viewers

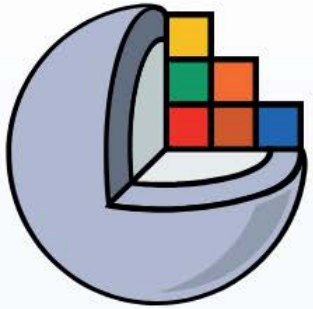




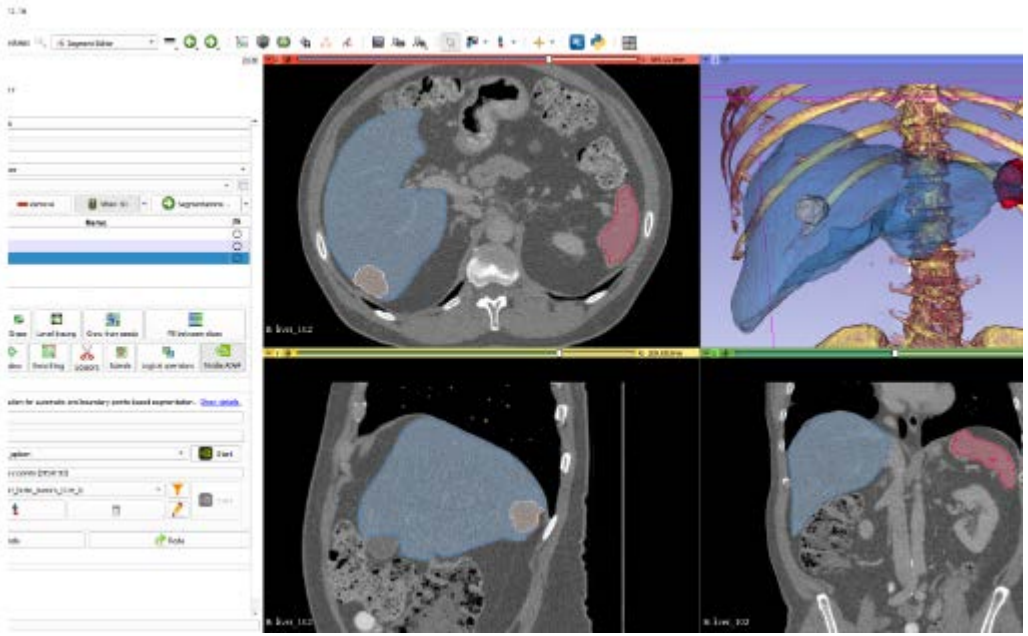


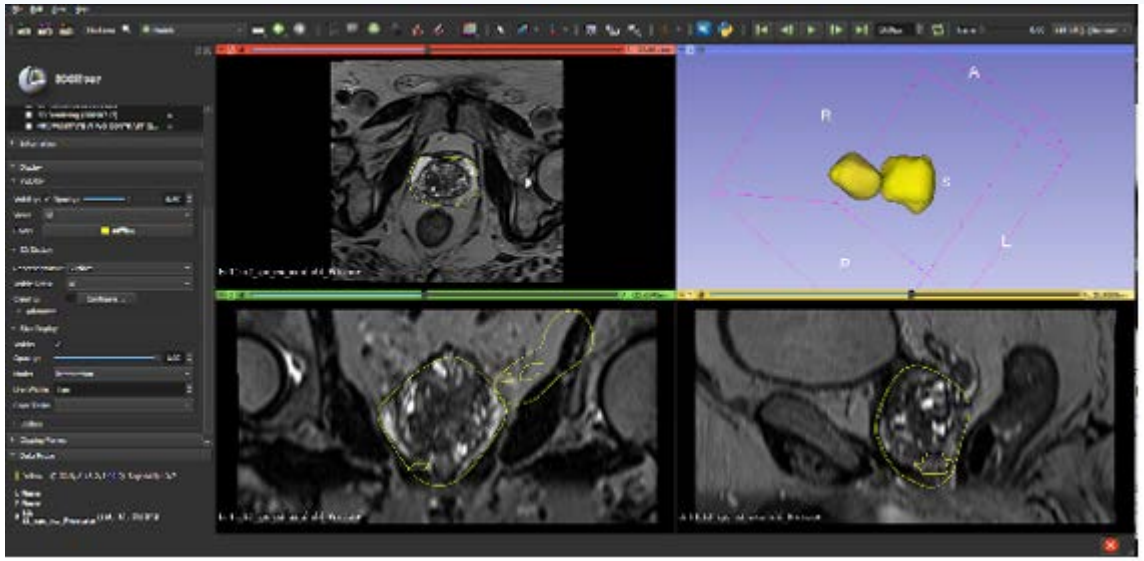
https://www.postdicom.com/en/blog/top-25-free-dicom-viewers

Image viewers



https://www.slicer.org/





3D Slicer image computing platform

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3D Slicer is a **free**, **open source** software for visualization, processing, segmentation, registration, and analysis of medical, biomedical, and other 3D images and meshes; and planning and navigating image-guided procedures.

https://www.youtube.com/watch?v=ZRYMItzwg8g&ab_channel=JanWitowski

